Jia-Wei Liu

Curriculum Vitae

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DOB: Jan. 24, 1996

Education

Sept. 2018 – Present **Tsinghua University**, *Master of Science*, Department of Engineering Mechanics, Institute of Dynamics and Control, Advisor: Prof. Ge-Xue Ren, Degree expected in July 2020.

Sept. 2014 – July 2018 **Northwestern Polytechnical University**, *Bachelor of Engineering*, Honors College. Overall GPA: 3.80/4.0, 91.84/100, Rank: 2/145, Major GPA: 3.88/4.0, 93.96/100.

Standardized Test

May 2019 TOEFL iBT: R28/L30/S21/W27, Total: 106.

Jan. 2019 GRE General: Verbal: 158(80%), Quantitative: 169(96%), AW: 4.0(59%), Total: 327+4.0.

Publications

June 2019 **Jia-Wei Liu**, Jia-Peng Liu, Xuan-Bo Shu, Aki Mikkola, Ge-Xue Ren, "An efficient multibody dynamic model of three-dimensional meshing contacts in helical gear-shaft system and its solution", Mechanism and Machine Theory, Under review.

Feb. 2019 **Jia-Wei Liu**, Jia-Peng Liu, Jia-Lun Yao, Ge-Xue Ren, "Dynamic characteristics of helical gear based on Arbitrary Lagrangian Eulerian formulation", in: ECCOMAS 2019 Multibody Dynamic Conference, Duisburg, Germany, 2019, Accepted.

Research Interests

Multibody dynamics, Gear system, ALE formulation, Vehicle system, Robotics.

Research Experience

Oct. 2018 – Present **Model order reduction method for simulation of planetary gear set based on Arbitrary Lagrangian Eulerian (ALE) formulation**, Research leader, Tsinghua University.

The planet gear is split into two ALE gears mentioned below so that each ALE planet gear only contacts with the ring or sun gear. The accurateness of replacing the planet gear with two ALE gears has been verified. The multibody dynamic modeling of the planetary gear set is under research, which can achieve great reduction of the degrees of freedom (DOFs) of the system.

Feb. 2019 – June 2019 **An efficient multibody dynamic modeling method of three-dimensional (3D) contacts in helical gear-shaft system**, Research leader, Tsinghua University.

The ALE helical gear was first proposed, where low frequency approximation (LFA) technique and ALE formulation were adopted to reduce the system DOFs from 10^6 to $10^3 \sim 10^4$. Then, inertial terms in dynamic equations were simplified and a high-efficiency contact algorithm based on hierarchical bounding box tree (OBB-tree) was proposed. Finally, 3D Timoshenko beam elements in Lagrangian formation were adopted to discretize the shaft. The proposed method was implemented using C++ and one order of magnitude acceleration of simulation time was achieved with high accuracy.

June 2018 – Feb. 2019 **Development of evaluation method for sliding wear by multibody dynamics**, Project leader, Tsinghua University, Partner: Mitsubishi Heavy Industries, Japan.

A general low-frequency approximated model was established using C++, which was successfully applied in the variable-geometry turbocharger (VGT) in this project to improve the computational efficiency. The hierarchical OBB-tree method was adopted for rapid contact detection in VGT.

Jan. 2018 – Oct. 2018 **Research on dynamic characteristics of helical gears and bevel gears based on ALE formulation**, Research leader, Tsinghua University.

Four pairs of helical gears with different helix angles and two pairs of bevel gears were established and meshed using Solidworks and Hypermesh, and the corresponding ALE gears were developed for higher efficiency. The dynamic transmission characteristics with correlation to three dimensional contacts and helix angles were analyzed, for which an algorithm to compute contact lines was proposed.

Work Experience

Jan. 2018 – Present **Research Assistant**, Multibody Dynamics Laboratory, Tsinghua University.

Contributed to the research of multibody dynamics of gear system based on the ALE formulation, including the helical gear pair, bevel gear pair, helical gear-shaft system and planetary gear set.

Sept. 2018 – Jan. 2019 **Teaching Assistant**, Theoretical Mechanics, Tsinghua University.

Provided 2 lectures on laboratory sessions for introductions of the theoretical mechanics. Offered homework correction twice a week and provided 5 Q&A sessions. Invigilated and graded on 3 exams.

Volunteers & Activities

Sept. 2018 – Jan. 2019 **Principle Member**, League Graduate, Tsinghua University.

Assisted in the design of a memorial video and offered advice on international students activities.

Sept. 2018 – Jan. 2019 **Principle Member**, Graduate Student Union, Tsinghua University.

Served as volunteer in the opening ceremony. Assisted in organizing academic lectures.

Apr. 2016 – June 2016 **Team Leader**, Project Management, Northwestern Polytechnical University.

Worked with teachers from the Netherlands and led a team to organize the route plannings in Africa.

Sept. 2015 – Sept. 2016 **Principle Member**, Enactus Practice Center, Northwestern Polytechnical University.

Collected and analyzed data for the water-shortage region and collaborated on inventing our first water filter equipment, which adopted soil infiltration and was about 5 times cheaper than the market.

Sept. 2014 – July 2018 **Class Leader**, Honors College, Northwestern Polytechnical University.

Offered supports for students' study and organized dozens of Q&A sessions and study lectures.

Honors & Awards

Jan. 2018 Scholarship of Innovation and Entrepreneurship in Honors College.

June 2017 Second Prize of C++ Programming in Northwestern Polytechnical University.

Mar. 2017 Third Prize of National English Contest for College Student.

2016 – 2017 Scholarship of First Prize in Honors College.

2016 – 2017 Scholarship of Outstanding Student in Northwestern Polytechnical University.

May 2016 Second Prize of Mathematical Modeling in Northwestern Polytechnical University.

2015 – 2016 Scholarship of First Prize in Honors College.

2015 – 2016 Scholarship of Outstanding Student in Northwestern Polytechnical University.

May 2015 Second Prize of Mathematical Modeling in Northwestern Polytechnical University.

2014 – 2015 Scholarship of First Prize in Honors College.

2014 – 2015 Scholarship of Outstanding Student in Northwestern Polytechnical University.

2014 – 2015 National Scholarship (Awarded to top 1% students in Honors College).

Skills & Abilities

Proficient in C/C++ Programming, Matlab, ABAQUS, Hypermesh, Solidworks, Adobe Illustrator, Latex(Lyx), MS Powerpoint, MS Word, MS Excel.